

Getting Started with **Git and GitHub** on Windows



Git is a popular distributed version control system that allows developers to manage their source code history. It is a great tool for collaboration between the several developers who work independently on a particular module, and helps track changes in code. GitHub is the online hosting service that hosts the Git repositories. This article is aimed at developers who use Windows as their primary operating system.

Software development is not about writing a piece of code at one go and compiling it into a binary, but is a process of making continuous and iterative changes.

Keeping track of things is often difficult when there are too many bits and pieces of code to deal with. This gets even more difficult when several versions of the code are required, just in case the developer wants to go back to a previous version if something is broken in the current one.

Software engineering is a job that involves considerable social interaction. Although developers may work alone at their desks, they usually work together in well separated modules to get the entire application running. Hence, they need to be able to collaborate and have a single working version while they all add their versions to the source code.

These requirements ushered in version control systems that allow developers do just that. Over several years, many version control systems have emerged, each with its own set of features and nuances. Some popular examples are Git, Subversion and Mercurial. These are widely used in

the software industry. This article looks at Git as a source control management system to keep track of code history.

Why Git?

Git is one of the most popular distributed version control systems, which allows developers to collaborate and keep track of the code history throughout the software development process. Its aim is speed, data integrity and distributed collaboration between non-linear workflows.

Git is open source and easy to get started with. It was originally authored by Linus Torvalds, the man behind Linux – one of the world's biggest open source collaboration projects. Today, it's a necessary skill in the software industry, as most companies rely on Git (or some other form of source control mechanism). While Git can be used to coordinate work among developers, it needs a centralised server to host the source code, dependencies and global history.

GitHub

GitHub is an online service that provides Git repository hosting and can be accessed from anywhere. It's the most popular Web based Git hosting

service with more than 40 million users. It hosts millions of open source and closed source repositories. GitHub has a simple and intuitive UI that allows users to get started with it easily without any prior knowledge of Git.

Let us look at two ways to use Git and GitHub to manage projects in Windows.

Creating a repository

Repositories are like folders for your projects and they contain your source code, configuration files, etc, which are stored in both GitHub servers and your local machine. GitHub allows you to create unlimited public and private repositories.

First, create an account at <https://github.com/> and verify your email address.

Then, to create a repository, click on the '+' button on the top navigation bar or go to <https://github.com/new>. The GitHub wizard will help you create your first repository.

Write a repository name that describes the purpose of this project. You can change it later. Then you can write an optional description of the project in the textbox below.